Angled launches

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use components to analyze an angled projectile.

SUCCESS CRITERIA - I can...

1. Separate the initial velocity into two components.
2. Use vertical motion to determine flight time.
3. Explain the assumptions behind range comparisons.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Downward velocity changes from 3 to 8 m/s in 0.5 s. Find acceleration and name one possible measurement uncertainty.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $t_{top} = v_{y,i}/g$; $t_{flight} = 2\,v_{y,i}/g$; range = $v_x\,t_{flight}$. Upward initial component $v_{y,i}$; g = 10 m/s². Equal launch/landing height and no air resistance; CPA components are supplied.

Worked reasoning: A projectile has initial components 4 m/s horizontal and 10 m/s upward. With g=10 m/s² it reaches the top in 1 s and returns to launch height in 2 s; range=8 m.

Your turn: Sketch its path and velocity arrows at launch, top and return.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

03 Read and interpret the model

At the highest point, which velocity component is zero and which remains? Is acceleration zero?



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

04 Apply the physics

A ball launches with velocity components 6 m/s horizontal and 10 m/s upward. It lands at its launch height. Use g = 10 m/s² and ignore air resistance. Label the model, then find time to the top, total flight time and range.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

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Continue down each column.

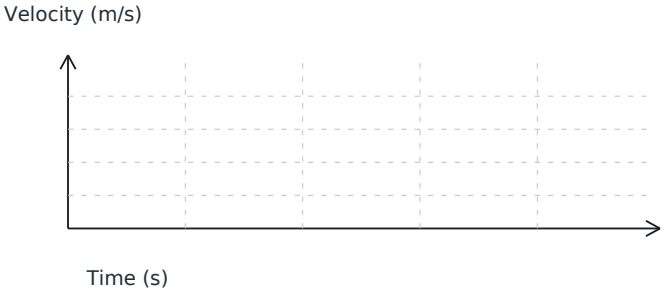
05 Explain your reasoning

Why is doubling time to the top valid only when the projectile returns to launch height in this ideal model?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 7 kg cart starts from rest under 63 N net force right for 6 s. Find acceleration and final velocity. State one assumption.
Label the given and predicted quantities on the model.



Marzano check - circle: 1 2 3

07 Exit check

Given v_x=4 m/s, initial v_y=15 m/s up, find time to top and total flight time to launch height. Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.

Separate horizontal and vertical motion

Label x and y; state positive directions.



Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2. Keep the upward component 10 m/s but double the horizontal component. Compare flight time, maximum height and range. Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3